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PowerCo

Hiring Team

Application: Master Thesis Regulatory Intelligence powered by AI in Product Management

Dear Hiring Team,

I am applying for the Master Thesis Regulatory Intelligence powered by AI in Product Management position at PowerCo in Salzgitter. The brief on designing a robust LLM based decision support system with agentic workflows, RAG, executive style reporting, and time series forecasting on heterogeneous data reads like a description of the two threads I have already been working on.

On the AI and agentic side, my Hybrid RAG Orchestrator runs on Llama 3.1 through Groq with LangChain, a stateful memory agent, ChromaDB retrieval, and a custom router that classifies user intent across local documents, DuckDuckGo web search, and direct conversation. That router is a working example of the multi agent routing and critique pattern the brief describes, and it gives me a starting point for a scalable system architecture with human oversight built in.

On the forecasting side, my current collaboration with eRay GmbH runs a recursive time series pipeline that benchmarks Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet on real sensor data, uses CatBoost multi quantile regression for asymmetric 80 percent intervals, and enforces strict anti leakage rules across the pipeline. That is exactly the classical machine learning to sequence modelling spectrum the brief calls for, applied to messy industrial signals.

The CreditIQ project adds the regulatory layer, a full EU AI Act Annex III and GDPR write up, model card, and a Streamlit decision support tool that keeps a human in the loop per GDPR Article 22. My Economic Impact of Global Climate Events analysis shows I can turn heterogeneous global event data into business relevant insight for stakeholders. Fluent English, German at B1 in progress. I would be glad to discuss the thesis scope in person or via video call.

Kind regards,

Rahul Rawat